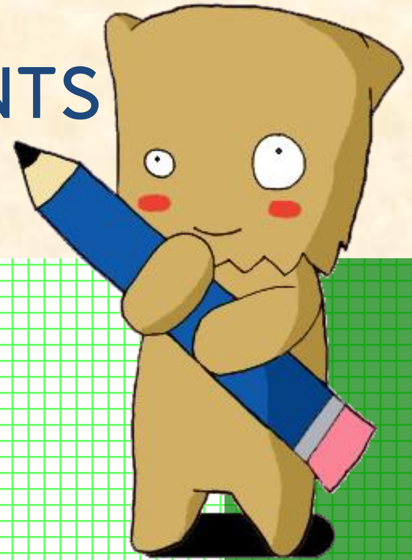
Chapter 7

DESIGN FOR ENGINEERING EXPERIMENTS



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Design for Engineering Experiments

- The Strategy of Experimentation
- Factorial Experiments
- 2^k Factorial Designs
- Fractional Replication of a 2^k Design

Introduction

- An experiment is a test or series of tests.
- The design of an experiment plays a major role in the eventual solution of the problem.
- In a factorial experimental design, experimental trials (or runs) are performed at all combinations of the factor levels.
- The analysis of variance (ANOVA) will be used as one of the primary tools for statistical data analysis.

The Strategy of Experimentation

Every experiment involves a sequence of activities:

1. Conjecture – the original hypothesis that motivates the experiment.

2. Experiment – the test performed to investigate the conjecture.

3. Analysis – the statistical analysis of the data from the experiment.

4. Conclusion – what has been learned about the original conjecture from the experiment. Often the experiment will lead to a revised conjecture, and a new experiment, and so forth.

Design for Engineering Experiments

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Factorial Experiments

Definition

By a **factorial experiment** we mean that in each complete trial or replicate of the experiment all possible combinations of the levels of the factors are investigated.

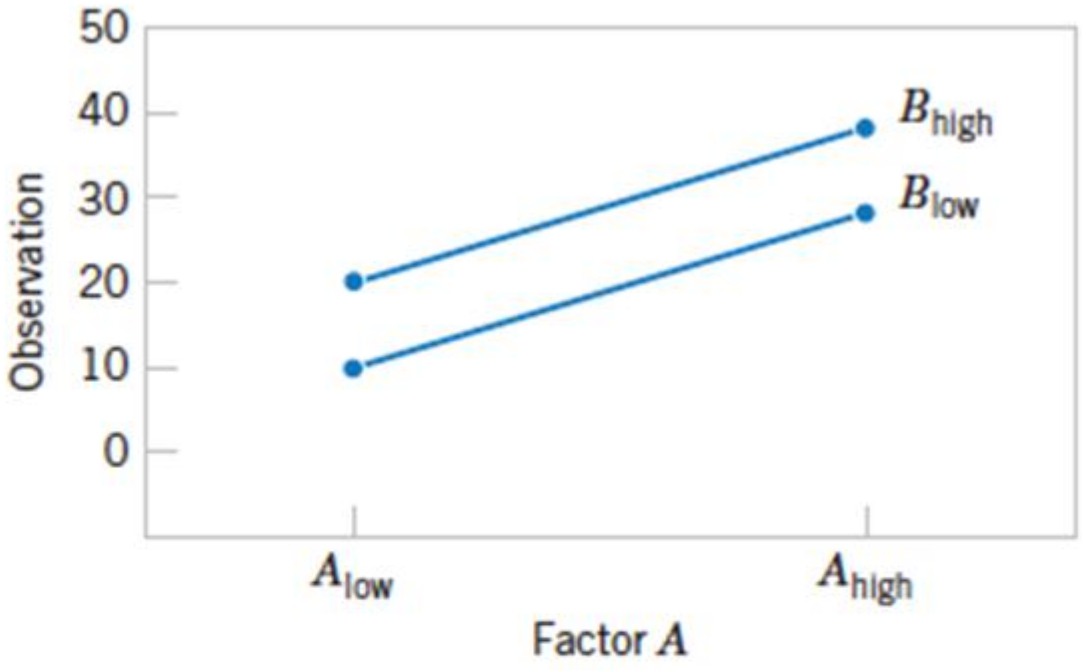
A Factorial Experiment with Two Factors

	Factor B	
Factor A	B_{low}	B_{high}
A_{low}	10	20
A_{high}	30	40

A Factorial Experiment with Interaction

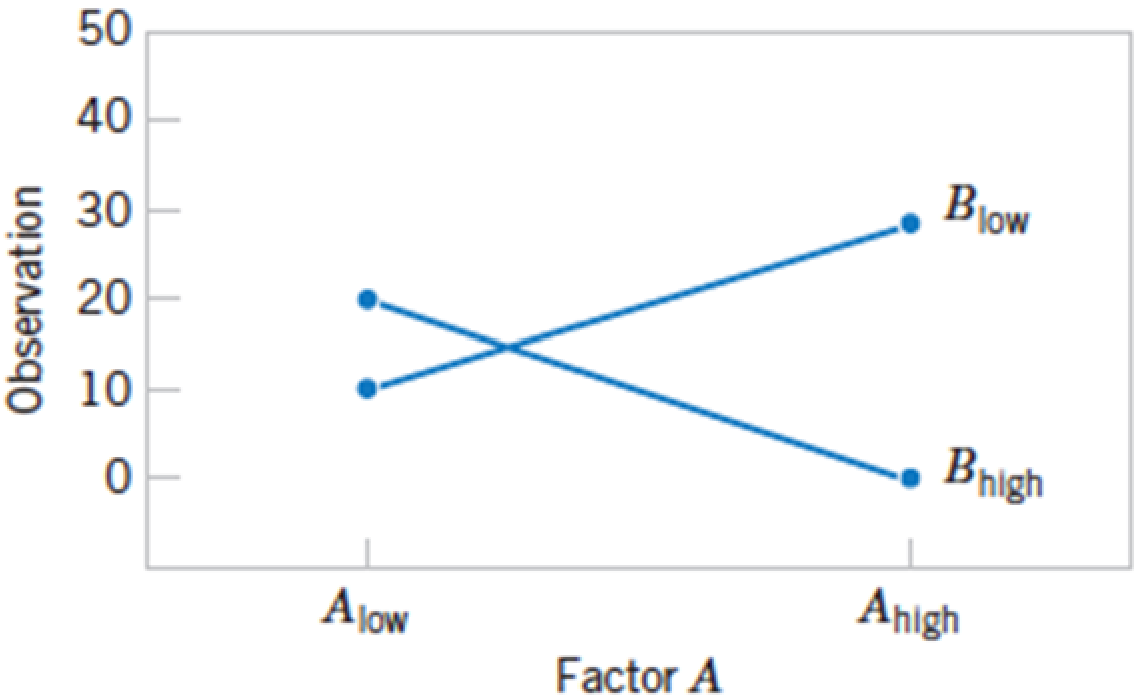
	Factor B	
Factor A	B_{low}	B_{high}
A_{low}	10	20
A_{high}	30	0

Factorial Experiments



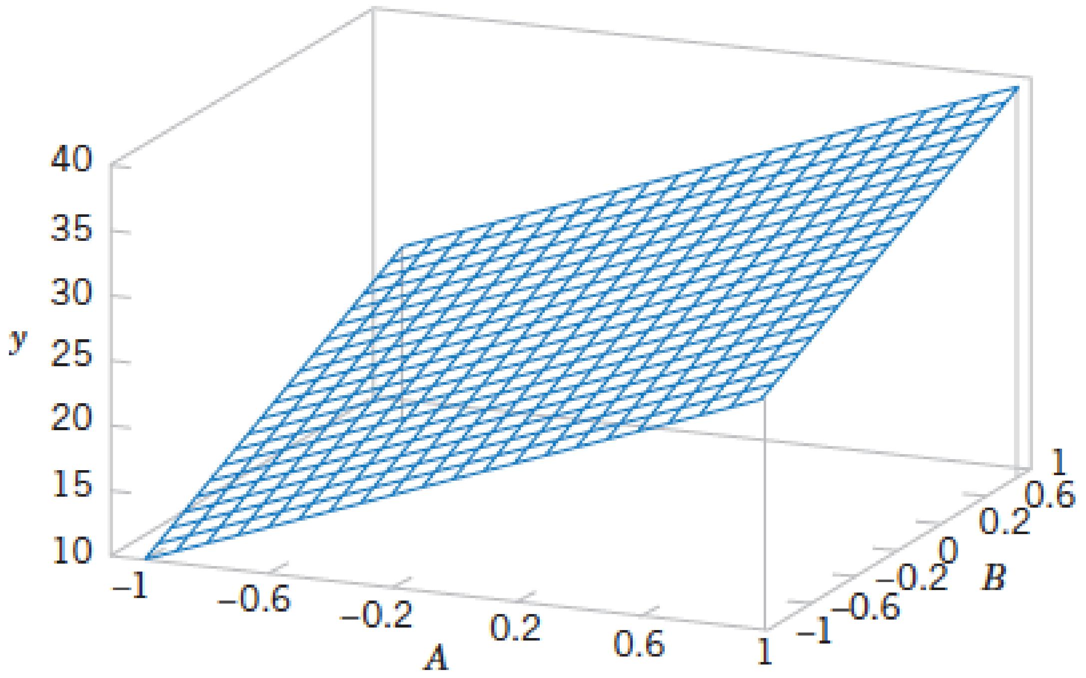
Factorial Experiment, no interaction.

Factorial Experiments



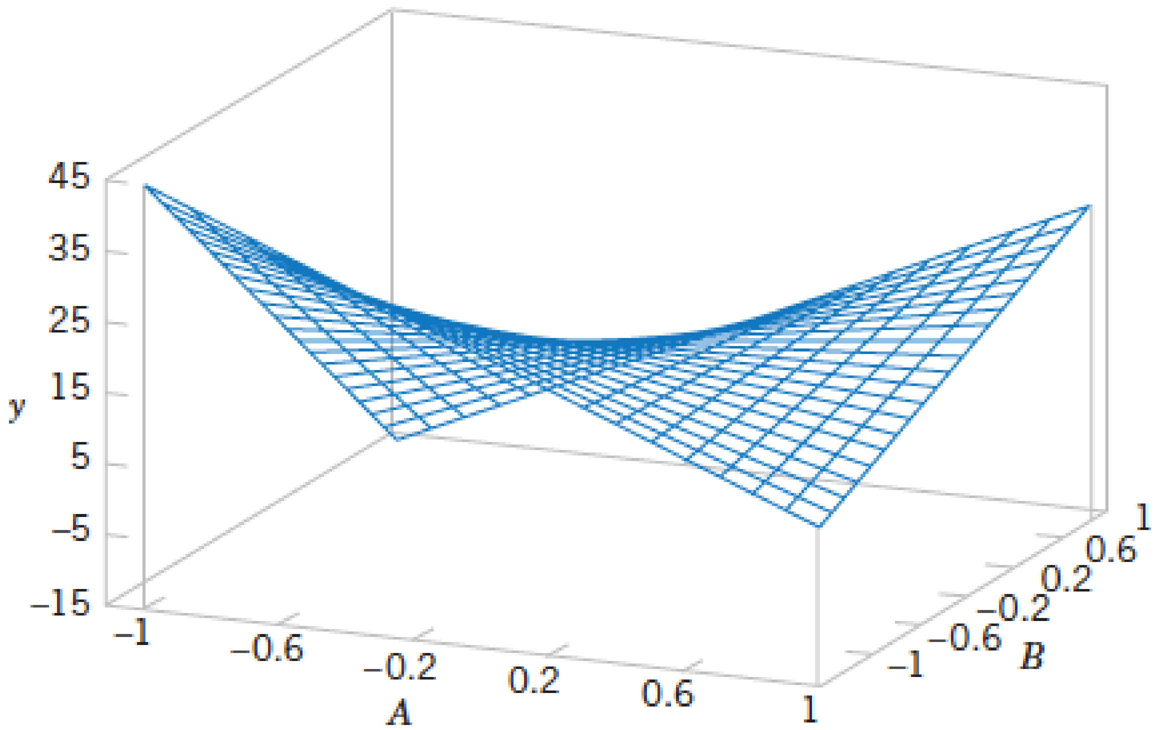
Factorial Experiment, with interaction.

Factorial Experiments



Three-dimensional surface plot of the data, showing main effects of the two factors *A* and *B*.

Factorial Experiments



Three–dimensional surface plot of the data, showing main effects of the *A* and *B* interaction.

Two-Factor Factorial Experiments

Data Arrangement for a Two-Factor Factorial Design

		Factor B					
		1	2	$\cdots$	b	Totals	Averages
	1	$y_{111}, y_{112}, \ldots, y_{11n}$	$y_{121}, y_{122}, \ldots, y_{12n}$		$y_{1b1}, y_{1b2}, \ldots, y_{1bn}$	$y_{1\cdot\cdot}$	$\bar{y}_{1\cdot\cdot}$
Factor A	2	$y_{211}, y_{212}, \ldots, y_{21n}$	$y_{221}, y_{222}, \ldots, y_{22n}$		$y_{2b1}, y_{2b2}, \ldots, y_{2bn}$	$y_{2\cdot\cdot}$	$\bar{y}_{2\cdot\cdot}$
	$\vdots$						
	a	$y_{a11}, y_{a12}, \ldots, y_{a1n}$	$y_{a21}, y_{a22}, \ldots, y_{a2n}$		$y_{ab1}, y_{ab2}, \ldots, y_{abn}$	$y_{a\cdot\cdot}$	$\bar{y}_{a\cdot\cdot}$
Totals		$y_{\cdot 1\cdot}$	$y_{\cdot 2\cdot}$		$y_{\cdot b\cdot}$	$y_{\cdots}$	
Averages		$\bar{y}_{\cdot 1\cdot}$	$\bar{y}_{\cdot 2\cdot}$		$\bar{y}_{\cdot b\cdot}$		$\bar{y}_{\cdots}$

Design for Engineering Experiments

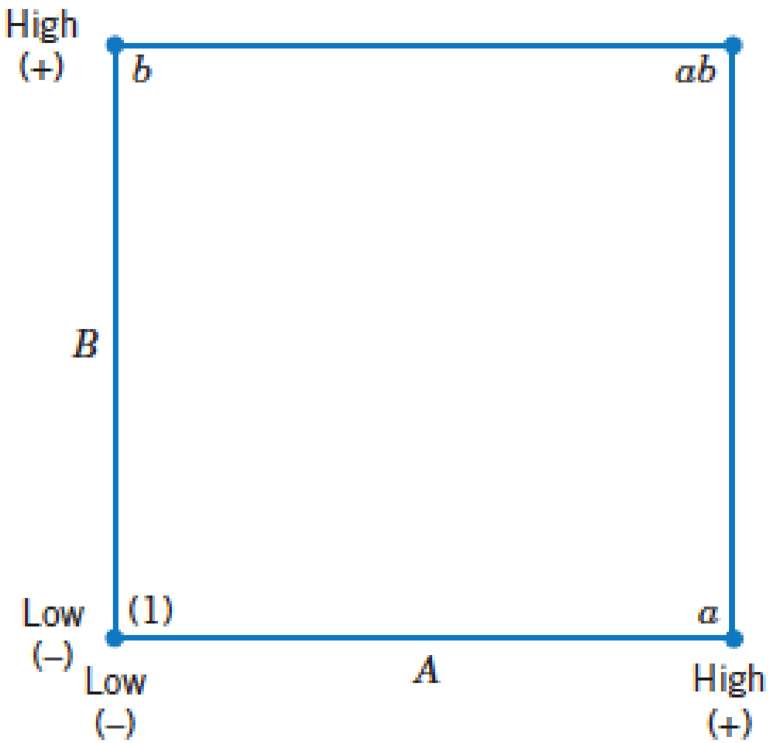
- The Strategy of Experimentation
- Factorial Experiments
- 2^k Factorial Designs
- Fractional Replication of a 2^k Design

2^k Factorial Designs

- Factorial designs are frequently used in experiments involving several factors where it is necessary to study the joint effect of the factors on a response.
- The most important of the special cased in that of *k* factors, each at only two levels.
- These levels may be quantitative, such as two values of temperature, pressure, or time; or they may be qualitative, such as two machines, two operators, the "high" and "low" levels of a factor, or perhaps the presence and absence of a factor. A complete replicate of such a design requires $2 \times 2 \times \ldots \times 2 = 2^k$ observations and is called a 2^k factorial design.

2^k Factorial Designs

- 2^2 Design



Treatment	A	B
(1)	−	−
a	+	−
b	−	+
ab	+	+

The 2^2 factorial design.

2^k Factorial Designs

2^2 Design

- The main effect of a factor A is estimated by

$$A = \bar{y}_{A+} - \bar{y}_{A-} = \frac{a + ab}{2n} - \frac{b + (1)}{2n} = \frac{1}{2n}[a + ab - b - (1)]$$

- The main effect of a factor B is estimated by

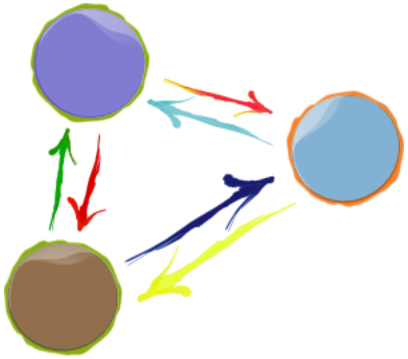
$$B = \bar{y}_{B+} - \bar{y}_{B-} = \frac{b + ab}{2n} - \frac{a + (1)}{2n} = \frac{1}{2n}[b + ab - a - (1)]$$

2^k Factorial Designs

2^2 Design

- The AB interaction effect is estimated by

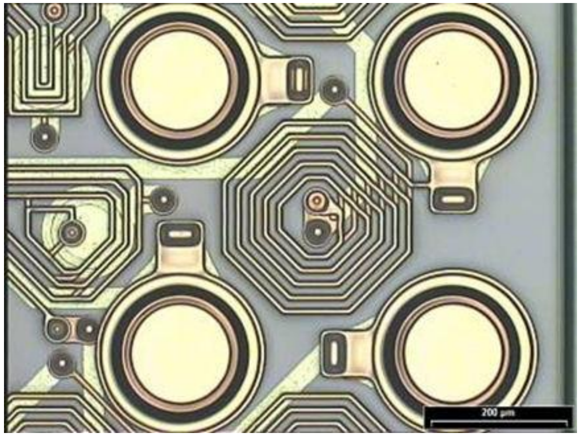
$$AB = \frac{ab + (1)}{2n} - \frac{a + b}{2n} = \frac{1}{2n}[ab + (1) - a - b]$$



Example: Wafer Layer

Wafer Layer

An article in the *AT&T Technical Journal* (Vol. 65, March/April 1986, pp. 39–50) describes the application of two-level factorial designs to integrated circuit manufacturing. A basic processing step in this industry is to grow an epitaxial layer on polished silicon wafers. The wafers are mounted on a susceptor and positioned inside a bell jar. Chemical vapors are introduced through nozzles near the top of the jar. The susceptor is rotated, and heat is applied. These conditions are maintained until the epitaxial layer is thick enough.



The 2^2 Design for the Epitaxial Process Experiment

Treatment Combination	Design Factors			Thickness (μm)				Thickness (μm)	
	A	*B*	*AB*					Total	Average
(1)	−	−	+	14.037	14.165	13.972	13.907	56.081	14.020
a	+	−	−	14.821	14.757	14.843	14.878	59.299	14.825
b	−	+	−	13.880	13.860	14.032	13.914	55.686	13.922
ab	+	+	+	14.888	14.921	14.415	14.932	59.156	14.789

2^k Factorial Designs

2^2 Design

- The quantities in brackets are called contrasts.
- For example, the *A* contrast is

$$\text{Contrast}_A = a + ab - b - (1)$$

Signs for Effects in the 2^2 Design

Treatment Combination	Factorial Effect *I*	*A*	*B*	*AB*
(1)	+	−	−	+
a	+	+	−	−
b	+	−	+	−
ab	+	+	+	+

2^k Factorial Designs

2^2 Design

- Contrasts are used in calculating both the effect estimates and the sums of squares for *A*, *B*, and the *AB* interaction. The sums of squares formulas are

$$SS_A = \frac{[a + ab - b - (1)]^2}{4n}$$

$$SS_B = \frac{[b + ab - a - (1)]^2}{4n}$$

$$SS_{AB} = \frac{[ab + (1) - a - b]^2}{4n}$$

Statistical Analysis

- Standard Error of the Effects

$$V(\text{Effect}) = \frac{\sigma^2}{N/2} + \frac{\sigma^2}{N/2} = \frac{2\sigma^2}{N/2} = \frac{\sigma^2}{n2^{k-2}}$$

$$\hat{\sigma}_i^2 = \frac{\sum_{j=1}^{n} (y_{ij} - \bar{y}_{i.})^2}{(n-1)} \qquad i = 1, 2, \ldots, 2^k$$

$$\hat{\sigma}^2 = \sum_{i=1}^{2^k} \frac{\hat{\sigma}_i^2}{2^k}$$

Statistical Analysis

- Standard Error of the Effects

Treatment Combination	Factorial Effect A	B	AB	Thickness (μm)				Thickness (mm) Total	Average	Variance
(1)	−	−	+	14.037	14.165	13.972	13.907	56.081	14.020	
a	+	−	−	14.821	14.757	14.843	14.878	59.299	14.825	
b	−	+	−	13.880	13.860	14.032	13.914	55.686	13.922	
ab	+	+	+	14.888	14.921	14.415	14.932	59.156	14.789	

To illustrate this approach for the epitaxial process experiment, we find that

$$\hat{\sigma}^2 = \frac{\qquad}{4} =$$

and the estimated standard error of each effect is

$$se(\text{Effect}) = \sqrt{[\hat{\sigma}^2/(n2^{k-2})]} = \sqrt{[0.0208/(4 \cdot 2^{2-2})]} = 0.072$$

Statistical Analysis

- Standard Error of the Effects
- The standard error and the resulting t ratio is compared to a t distribution with 12 degree of freedom. The t ratio is used to judge whether the effect is significantly different from zero. Two standard error limits on the effect estimations are approximate 95% CI.

Effect	Effect Estimate	Estimated Standard Error	t Ratio	P-Value	Effect ± Two Estimated Standard Errors
A	0.836	0.072	11.61	0.00	0.836 ± 0.144
B	−0.067	0.072	−0.93	0.38	-0.067 ± 0.144
AB	0.032	0.072	0.44	0.67	0.032 ± 0.144

Degrees of freedom $= 2^k(n-1) = 2^2(4-1) = 12$.

Statistical Analysis

- Regression analysis

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \epsilon$$

- The least squares fitted model is

$$\hat{y} = 14.389 + \left(\frac{0.836}{2}\right) x_1 + \left(\frac{-0.067}{2}\right) x_2 + \left(\frac{0.032}{2}\right) x_1 x_2$$

$$\hat{y} = 14.389 + \left(\frac{0.836}{2}\right) x_1$$

Residual Analysis and Model Checking

$$\hat{y} = 14.389 + \left(\frac{0.836}{2}\right)x_1$$

- when $x_1 = -1$:

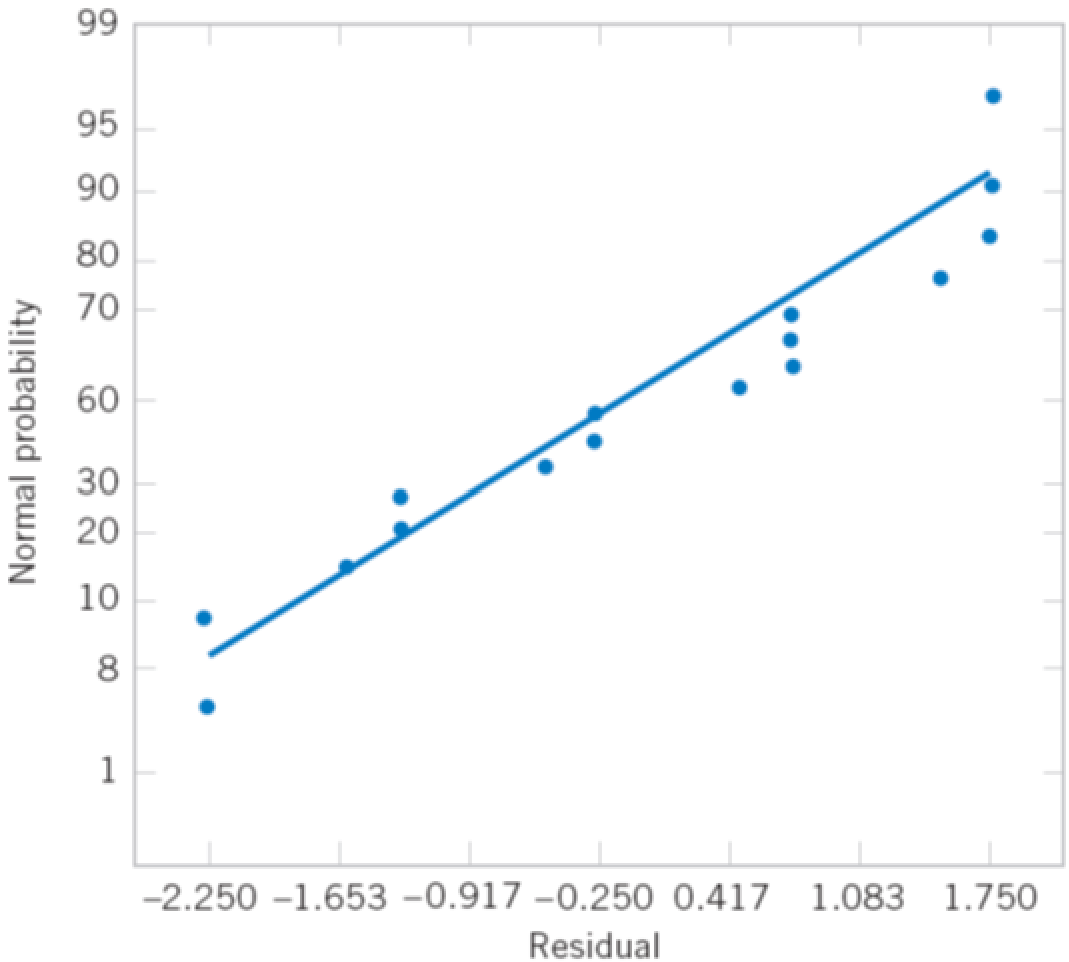
$$\hat{y} = 14.389 + \left(\frac{0.836}{2}\right)(-1) = 13.971\ \mu\text{m}$$

- The resulting residuals are:

$$e_1 = 14.037 - 13.971 =$$
$$e_2 = 14.165 - 13.971 =$$
$$e_3 = 13.972 - 13.971 =$$
$$e_4 = 13.907 - 13.971 =$$

Residual Analysis and Model Checking

Normal probability plot of residuals for the epitaxial process experiment.



2^k Factorial Designs

Formulas for Two-Level Factorial Experiments with k Factors Each at Two Levels and N Total Trials

$$\text{Coefficient} = \frac{\text{effect}}{2}$$

$$se(\text{Effect}) = \sqrt{\frac{2\hat{\sigma}^2}{N/2}} = \sqrt{\frac{\hat{\sigma}^2}{n2^{k-2}}}$$

$$se(\text{Coefficient}) = \frac{1}{2}\sqrt{\frac{2\hat{\sigma}^2}{N/2}} = \frac{1}{2}\sqrt{\frac{\hat{\sigma}^2}{n2^{k-2}}}$$

$$t \text{ ratio} = \frac{\text{effect}}{se(\text{effect})} = \frac{\text{coefficient}}{se(\text{coefficient})}$$

$$\hat{\sigma}^2 = \text{mean square error}$$

$$2^k(n-1) = \text{residual degrees of freedom}$$

P–value

The ***P*-value** is the smallest level of significance that would lead to rejection of the null hypothesis H_0 with the given data.

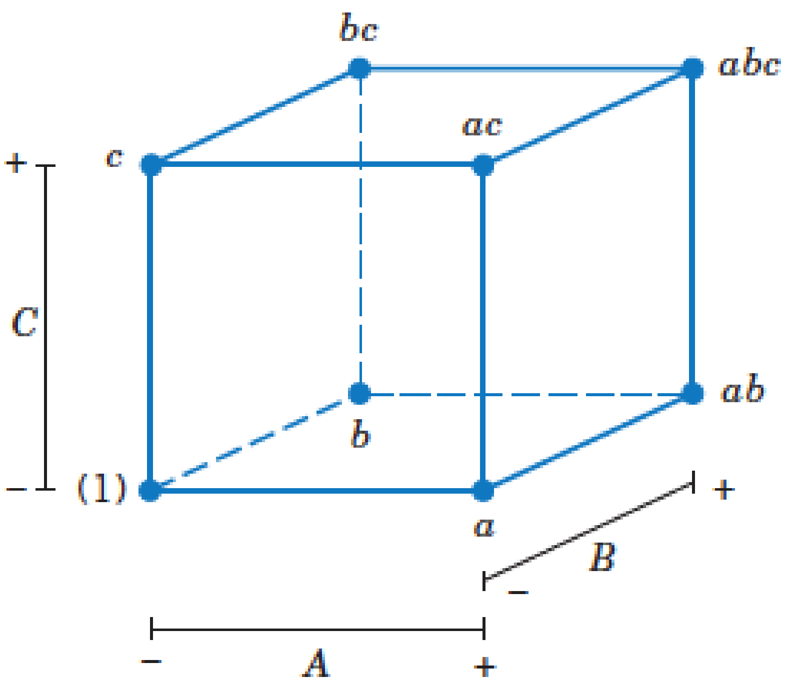
$$P = \begin{cases} 2[1 - \Phi(|t_0|)] & \text{for a two-tailed test: } H_0\text{: } \mu = \mu_0 \quad H_1\text{: } \mu \neq \mu_0 \\ 1 - \Phi(t_0) & \text{for a upper-tailed test: } H_0\text{: } \mu = \mu_0 \quad H_1\text{: } \mu > \mu_0 \\ \Phi(t_0) & \text{for a lower-tailed test: } H_0\text{: } \mu = \mu_0 \quad H_1\text{: } \mu < \mu_0 \end{cases}$$

Effect	Effect Estimate	Estimated Standard Error	t Ratio	P-Value	Effect ± Two Estimated Standard Errors
A	0.836	0.072	11.61	0.00	0.836 ± 0.144
B	−0.067	0.072	−0.93	0.38	−0.067 ± 0.144
AB	0.032	0.072	0.44	0.67	0.032 ± 0.144

Degrees of freedom $= 2^k(n - 1) = 2^2(4 - 1) = 12$.

2^k Factorial Designs

2^k Design for $k \geq 3$ Factors



(a) Geometric view

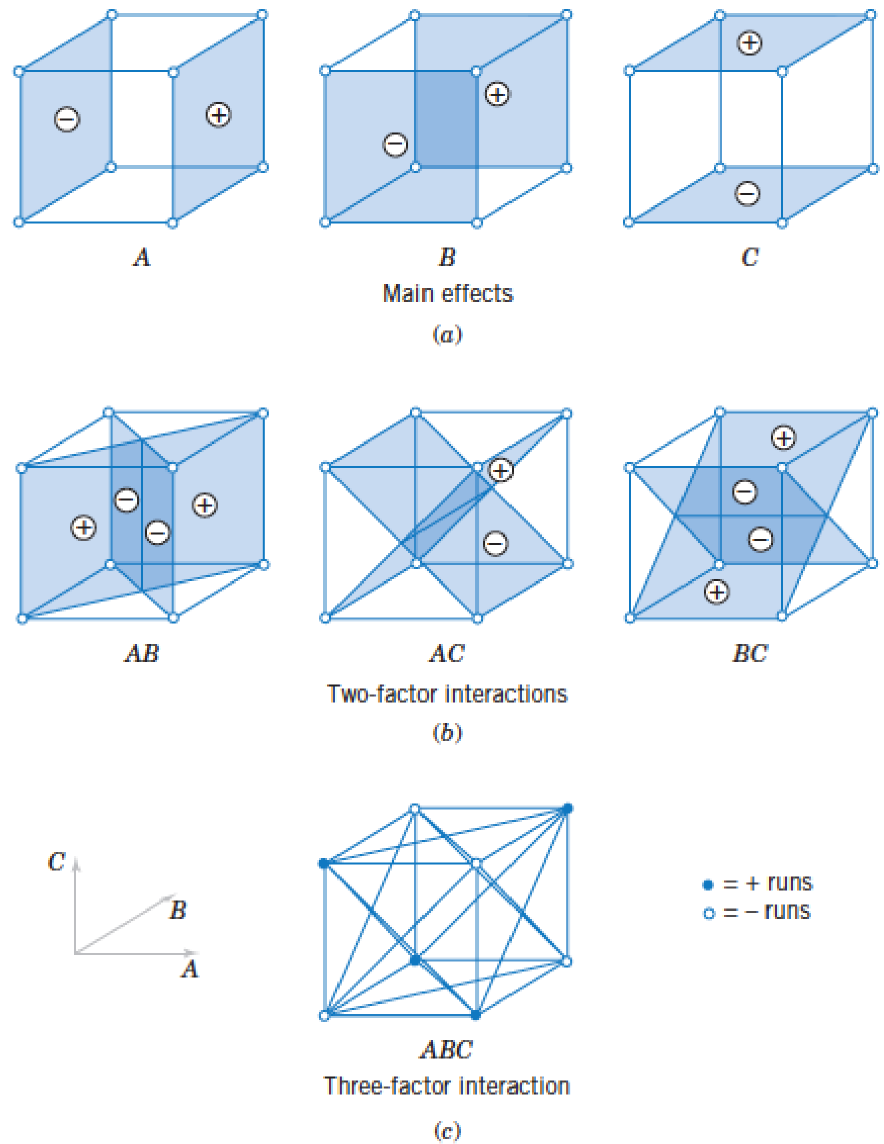
Run	A	B	C
1	−	−	−
2	+	−	−
3	−	+	−
4	+	+	−
5	−	−	+
6	+	−	+
7	−	+	+
8	+	+	+

(b) The 2^3 design matrix

The 2^3 design.

2^k Factorial Designs

2^k Design for $k \geq 3$ Factors



Geometric presentation of contrasts corresponding to the main effects and interaction in the 2^3 design. (a) Main effects. (b) Two–factor interactions. (c) Three–factor interaction.

2^k Factorial Designs

2^k Design for k ≥ 3 Factors

- The main effect of *A* is estimated by

$$A = \bar{y}_{A+} - \bar{y}_{A-}$$
$$= \frac{1}{4n}[a + ab + ac + abc - (1) - b - c - bc]$$

- The main effect of *B* is estimated by

$$B = \bar{y}_{B+} - \bar{y}_{B-}$$
$$= \frac{1}{4n}[b + ab + bc + abc - (1) - a - c - ac]$$

2^k Factorial Designs

2^k Design for k ≥ 3 Factors

- The main effect of *C* is estimated by

$$C = \bar{y}_{C+} - \bar{y}_{C-} = \frac{1}{4n}\left[c + ac + bc + abc - (1) - a - b - ab\right]$$

- The interaction effect of *AB* is estimated by

$$AB = \frac{1}{4n}\left[abc - bc + ab - b - ac + c - a + (1)\right]$$

2^k Factorial Designs

2^k Design for k ≥ 3 Factors

- Other two–factor interactions effects estimated by

$$AC = \frac{1}{4n}[(1) - a + b - ab - c + ac - bc + abc]$$

$$BC = \frac{1}{4n}[(1) + a - b - ab - c - ac + bc + abc]$$

- The three–factor interaction effect, *ABC,* is estimated by

$$ABC = \frac{1}{4n}[abc - bc - ac + c - ab + b + a - (1)]$$

2^k Factorial Designs

2^k Design for $k \geq 3$ Factors

Algebraic Signs for Calculating Effects in the 2^3 Design

Treatment Combination	Factorial Effect							
	I	A	B	AB	C	AC	BC	ABC
(1)	+	−	−	+	−	+	+	−
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
ab	+	+	+	+	−	−	−	−
c	+	−	−	+	+	−	−	+
ac	+	+	−	−	+	+	−	−
bc	+	−	+	−	+	−	+	−
abc	+	+	+	+	+	+	+	+

2^k Factorial Designs

2^k Design for k ≥ 3 Factors

- There are several interesting properties:

1. Except for the identity column I, each column has an equal number of plus and minus signs.
2. The sum of products of signs in any two columns is zero; that is, the columns in the table are **orthogonal.**
3. Multiplying any column by column I leaves the column unchanged; that is, I is an **identity element.**
4. The product of any two columns yields a column in the table, for example $A \times B = AB$, and $AB \times ABC = A^2B^2C = C$, since any column multiplied by itself is the identity column.

2^k Factorial Designs

2^k Design for k ≥ 3 Factors

- Contrasts can be used to calculate several quantities:

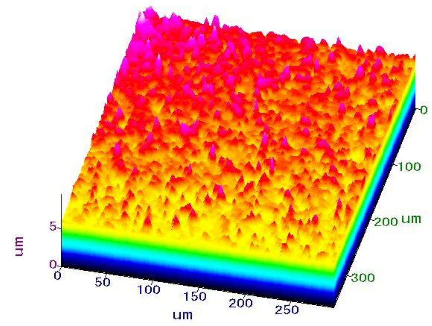
$$\text{Effect} = \frac{\text{Contrast}}{n2^{k-1}}$$

$$SS = \frac{(\text{Contrast})^2}{n2^k}$$

Example: Surface Roughness

Surface Roughness

Consider the surface roughness experiment originally described in Example 14-2. This is a 2^3 factorial design in the factors feed rate (A), depth of cut (B), and tool angle (C), with $n = 2$ replicates.



Surface Roughness Data

Treatment Combinations	Design Factors A	B	C	Surface Roughness	Totals
(1)	−1	−1	−1	9, 7	16
a	1	−1	−1	10, 12	22
b	−1	1	−1	9, 11	20
ab	1	1	−1	12, 15	27
c	−1	−1	1	11, 10	21
ac	1	−1	1	10, 13	23
bc	−1	1	1	10, 8	18
abc	1	1	1	16, 14	30

Example: Surface Roughness

The effect of A, for example, is

$$\begin{aligned} A &= \frac{1}{4n}[a + ab + ac + abc - (1) - b - c - bc] \\ &= \frac{1}{4(2)}[22 + 27 + 23 + 40 - 16 - 20 - 21 - 18] \\ &= \frac{1}{8}[27] = \end{aligned}$$

and the sum of squares for A is found :

$$SS_A = \frac{(\text{Contrast}_A)^2}{n2^k} = \frac{(27)^2}{2(8)} =$$

It is easy to verify that the other effects are

$$\begin{aligned} B &= \\ C &= \\ AB &= \\ AC &= \\ BC &= \\ ABC &= \end{aligned}$$

Examining the magnitude of the effects clearly shows that feed rate (factor A) is dominant, followed by depth of cut (B) and the AB interaction, although the interaction effect is relatively small.

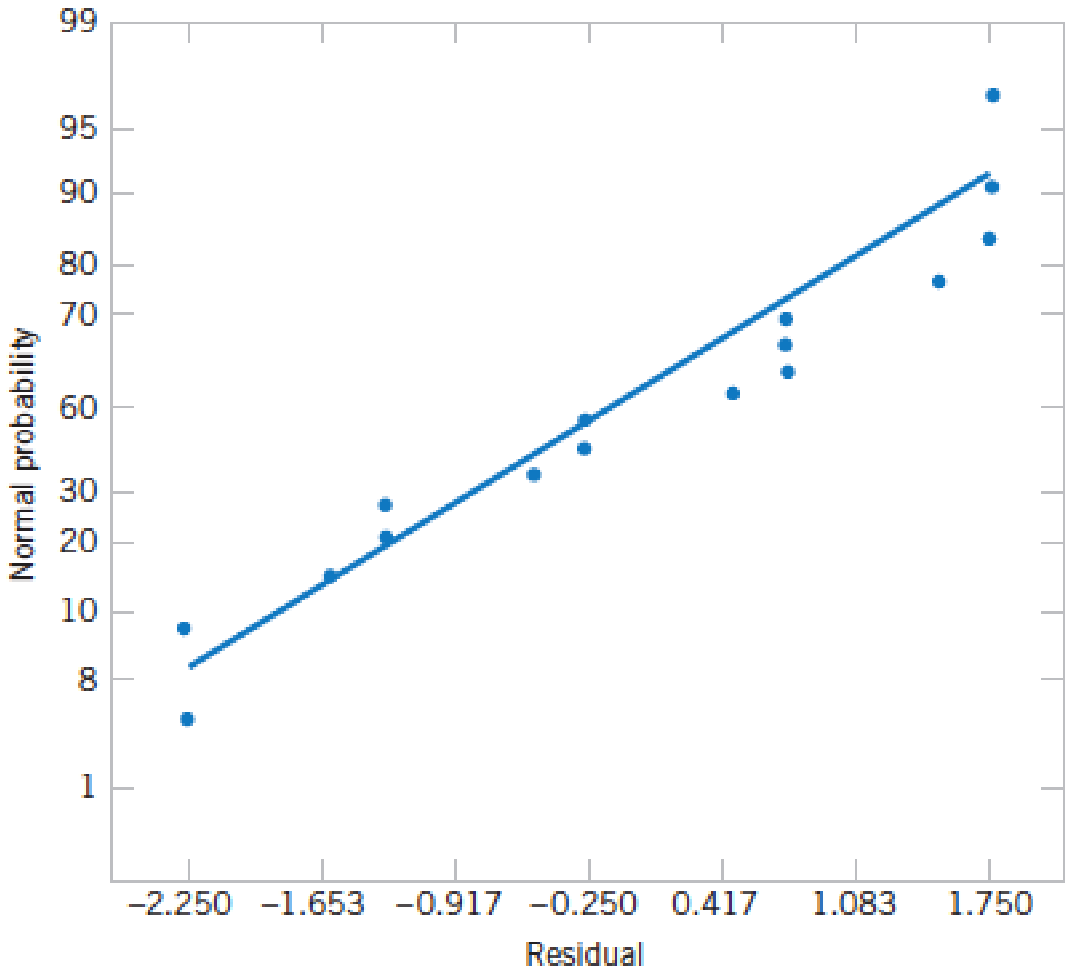
Regression Model and Residual Analysis

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \epsilon$$

$$\hat{y} = 11.0625 + \left(\frac{3.375}{2}\right)x_1 + \left(\frac{1.625}{2}\right)x_2 + \left(\frac{1.375}{2}\right)x_1 x_2$$

$$\hat{y} = 11.0625 + \left(\frac{3.375}{2}\right)(-1) + \left(\frac{1.625}{2}\right)(-1) + \left(\frac{1.375}{2}\right)(-1)(-1)$$
$$= 9.25$$

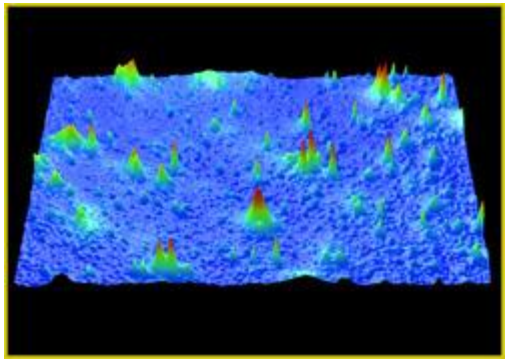
Residual Analysis



Example: Surface Roughness

Analysis of Variance for the Surface Finish Experiment

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	f_0	P-Value
A	45.5625	1	45.5625	18.69	0.0025
B	10.5625	1	10.5625	4.33	0.0709
C	3.0625	1	3.0625	1.26	0.2948
AB	7.5625	1	7.5625	3.10	0.1162
AC	0.0625	1	0.0625	0.03	0.8784
BC	1.5625	1	1.5625	0.64	0.4548
ABC	5.0625	1	5.0625	2.08	0.1875
Error	19.5000	8	2.4375		
Total	92.9375	15			

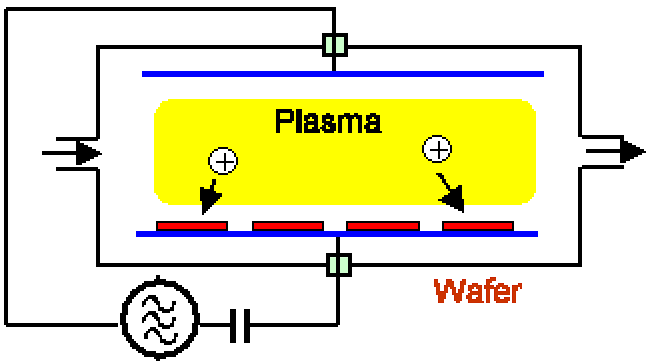


Single Replicate of the 2^k Design

Plasma Etch

An article in *Solid State Technology* ["Orthogonal Design for Process Optimization and Its Application in Plasma Etching" (May 1987, pp. 127–132)] describes the application of factorial designs in developing a nitride etch process on a single-wafer plasma etcher. The process uses C_2F_6 as the reactant gas. It is possible to vary the gas flow, the power applied to the cathode, the pressure in the reactor chamber, and the spacing between the anode and the cathode (gap). Several response variables would usually be of interest in this process, but in this example we will concentrate on etch rate for silicon nitride.

We will use a single replicate of a 2^4 design to investigate this process. Since it is unlikely that the three- and four-factor interactions are significant, we will tentatively plan to combine them as an estimate of error. The factor levels used in the design are shown below:



Design Factor

Level	Gap (cm)	Pressure (mTorr)	C_2F_6 Flow (SCCM)	Power (w)
Low (−)	0.80	450	125	275
High (+)	1.20	550	200	325

Single Replicate of the 2^k Design

The signs in the columns of this table can be used to estimate the factor effects. For example, the estimate of factor A is

$$\begin{aligned} A &= \frac{1}{8}[a + ab + ac + abc + ad + abd + acd \\ &\quad + abcd - (1) - b - c - bc - d - bd - cd - bcd] \\ &= \frac{1}{8}[669 + 650 + 642 + 635 + 749 + 868 + 860 \\ &\quad + 729 - 550 - 604 - 633 - 601 - 1037 \\ &\quad - 1052 - 1075 - 1063] \\ &= -101.625 \end{aligned}$$

Single Replicate of the 2^k Design

Contrast Constants for the 2^4 Design

	A	*B*	*AB*	*C*	*AC*	*BC*	*ABC*	*D*	*AD*	*BD*	*ABD*	*CD*	*ACD*	*BCD*	*ABCD*
(1)	−	−	+	−	+	+	−	−	+	+	−	+	−	−	+
a	+	−	−	−	−	+	+	−	−	+	+	+	+	−	−
b	−	+	−	−	+	−	+	−	+	−	+	+	−	+	−
ab	+	+	+	−	−	−	−	−	−	−	−	+	+	+	+
c	−	−	+	+	−	−	+	−	+	+	−	−	+	+	−
ac	+	−	−	+	+	−	−	−	−	+	+	−	−	+	+
bc	−	+	−	+	−	+	−	−	+	−	+	−	+	−	+
abc	+	+	+	+	+	+	+	−	−	−	−	−	−	−	−
d	−	−	+	−	+	+	−	+	−	−	+	−	+	+	−
ad	+	−	−	−	−	+	+	+	+	−	−	−	−	+	+
bd	−	+	−	−	+	−	+	+	−	+	−	−	+	−	+
abd	+	+	+	−	−	−	−	+	+	+	+	−	−	−	−
cd	−	−	+	+	−	−	+	+	−	−	+	+	−	−	+
acd	+	−	−	+	+	−	−	+	+	−	−	+	+	−	−
bcd	−	+	−	+	−	+	−	+	−	+	−	+	−	+	−
abcd	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+

Single Replicate of the 2^k Design

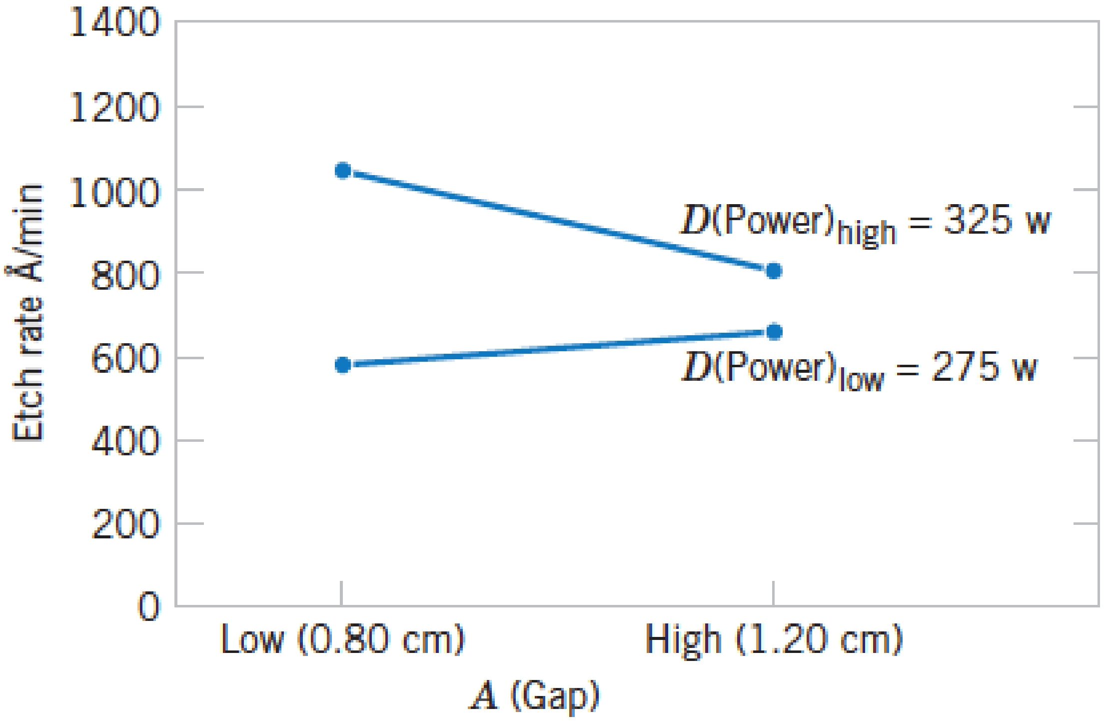
$$
\begin{aligned}
A &= & \qquad AD &= \\
B &= & \qquad BD &= \\
AB &= & \qquad ABD &= \\
C &= & \qquad CD &= \\
AC &= & \qquad ACD &= \\
BC &= & \qquad BCD &= \\
ABC &= & \qquad ABCD &= \\
D &= & &
\end{aligned}
$$

- Clearly, the main effects of A and D and the AD interaction are significant.

The residuals from the experiment in Example can be obtained from the regression model

$$\hat{y} = 776.0625 - \left(\frac{101.625}{2}\right) x_1 + \left(\frac{306.125}{2}\right) x_4 - \left(\frac{153.625}{2}\right) x_1 x_4$$

Single Replicate of the 2^k Design



AD (Gap–Power) interaction from the plasma etch experiment.

Analysis of variance

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	f_0	*P*-Value
A	41,310.563	1	41,310.563	20.28	0.0064
B	10.563	1	10.563	<1	—
C	217.563	1	217.563	<1	—
D	374,850.063	1	374,850.063	183.99	0.0000
AB	248.063	1	248.063	<1	—
AC	2,475.063	1	2,475.063	1.21	0.3206
AD	94,402.563	1	94,402.563	46.34	0.0010
BC	7,700.063	1	7,700.063	3.78	0.1095
BD	1.563	1	1.563	<1	—
CD	18.063	1	18.063	<1	—
Error	10,186.813	5	2,037.363		
Total	531,420.938	15			

$$F_0 = \frac{SS_{\text{Treatments}}/(a-1)}{SS_E/[a(n-1)]} = \frac{MS_{\text{Treatments}}}{MS_E}$$

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	Expected Mean Squares	F_0
A	SS_A	$a-1$	MS_A	$\sigma^2 + \dfrac{bcn\sum\tau_i^2}{a-1}$	$\dfrac{MS_A}{MS_E}$
B	SS_B	$b-1$	MS_B	$\sigma^2 + \dfrac{acn\sum\beta_j^2}{b-1}$	$\dfrac{MS_B}{MS_E}$
C	SS_C	$c-1$	MS_C	$\sigma^2 + \dfrac{abn\sum\gamma_k^2}{c-1}$	$\dfrac{MS_C}{MS_E}$
AB	SS_{AB}	$(a-1)(b-1)$	MS_{AB}	$\sigma^2 + \dfrac{cn\sum\sum(\tau\beta)_{ij}^2}{(a-1)(b-1)}$	$\dfrac{MS_{AB}}{MS_E}$
AC	SS_{AC}	$(a-1)(c-1)$	MS_{AC}	$\sigma^2 + \dfrac{bn\sum\sum(\tau\gamma)_{ik}^2}{(a-1)(c-1)}$	$\dfrac{MS_{AC}}{MS_E}$
BC	SS_{BC}	$(b-1)(c-1)$	MS_{BC}	$\sigma^2 + \dfrac{an\sum\sum(\beta\gamma)_{jk}^2}{(b-1)(c-1)}$	$\dfrac{MS_{BC}}{MS_E}$
ABC	SS_{ABC}	$(a-1)(b-1)(c-1)$	MS_{ABC}	$\sigma^2 + \dfrac{n\sum\sum\sum(\tau\beta\gamma)_{ijk}^2}{(a-1)(b-1)(c-1)}$	$\dfrac{MS_{ABC}}{MS_E}$
Error	SS_E	$abc(n-1)$	MS_E	σ^2	
Total	SS_T	$abcn-1$			

Design for Engineering Experiments

- The Strategy of Experimentation
- Factorial Experiments
- 2^k Factorial Designs
- Fractional Replication of a 2^k Design

Fractional Replication of the 2^k Design

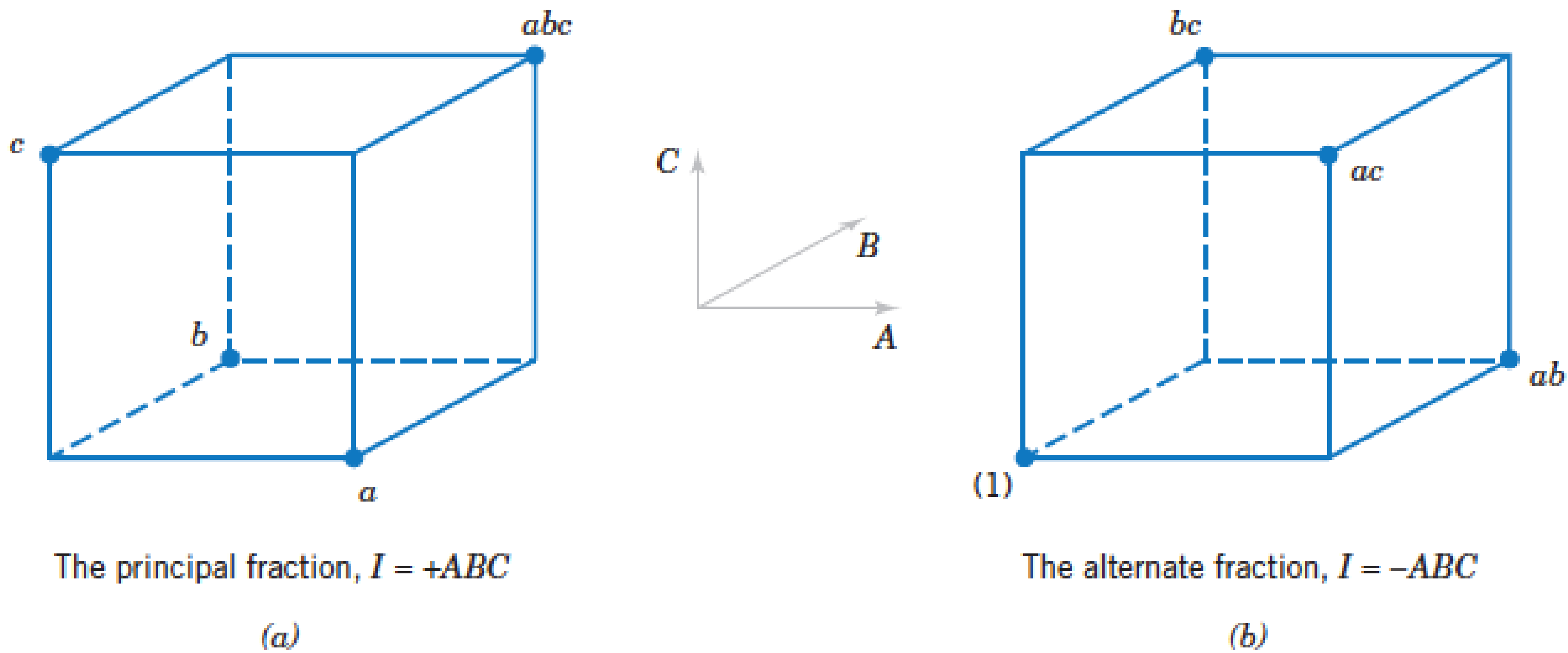
One–Half Fraction of the 2^k Design

Plus and Minus Signs for the 2^3 Factorial Design

Treatment Combination	Factorial Effect							
	I	A	B	C	AB	AC	BC	ABC
a	+	+	−	−	−	−	+	+
b	+	−	+	−	−	+	−	+
c	+	−	−	+	+	−	−	+
abc	+	+	+	+	+	+	+	+
ab	+	+	+	−	+	−	−	−
ac	+	+	−	+	−	+	−	−
bc	+	−	+	+	−	−	+	−
(1)	+	−	−	−	+	+	+	−

Fractional Replication of the 2^k Design

One–Half Fraction of the 2^k Design



The one–half fractions of the 2^3 design.

(a) The principal fraction, I = +*ABC*. (B) The alternate fraction, I = −*ABC*

Example: Plasma Etch

Plasma Etch

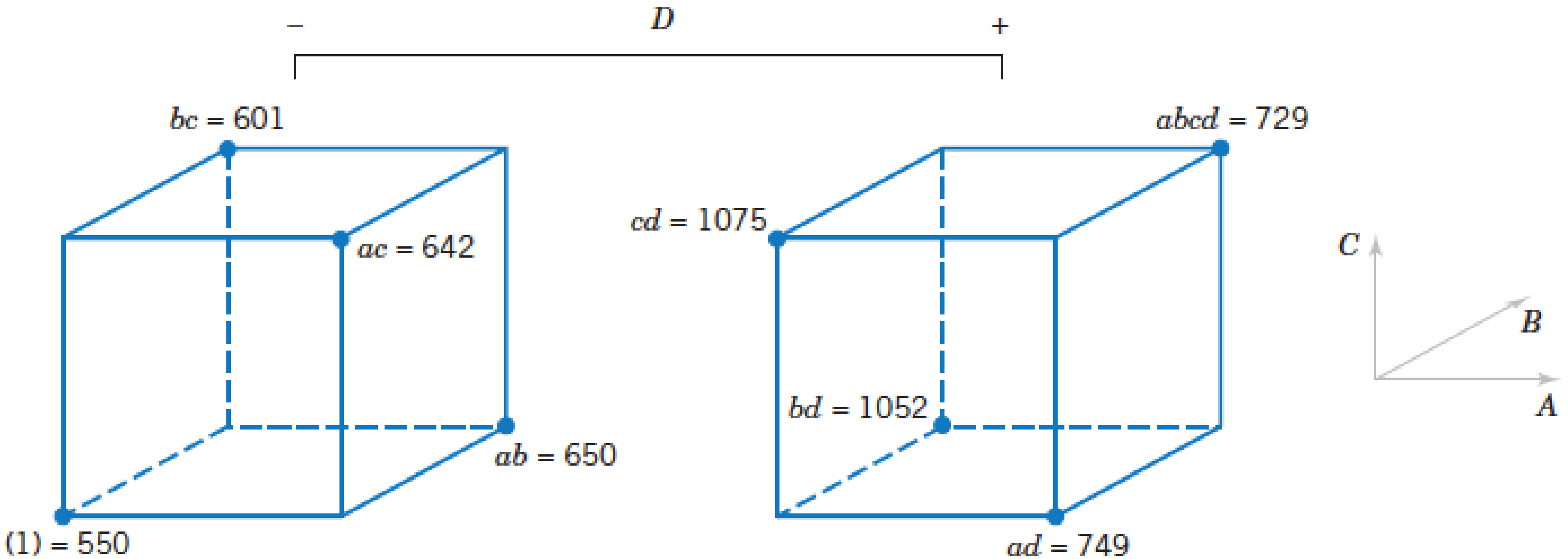
To illustrate the use of a one-half fraction, consider the plasma etch experiment. Suppose that we decide to use a 2^{4-1} design with $I = ABCD$ to investigate the four factors gap (A), pressure (B), C_2F_6 flow rate (C), and power setting (D).

This design would be constructed by writing down as the basic design a 2^3 in the factors A, B, and C and then setting the levels of the fourth factor $D = ABC$.

The 2^{4-1} Design with Defining Relation $I = ABCD$

A	B	C	$D = ABC$	Treatment Combination	Etch Rate
−	−	−	−	(1)	550
+	−	−	+	*ad*	749
−	+	−	+	*bd*	1052
+	+	−	−	*ab*	650
−	−	+	+	*cd*	1075
+	−	+	−	*ac*	642
−	+	+	−	*bc*	601
+	+	+	+	*abcd*	729

Example: Plasma Etch



The 2^{4-1} design for the experiment.

Example: Plasma Etch

In this design, the main effects are aliased with the three-factor interactions; note that the alias of A is

$$A \cdot I = A \cdot ABCD \qquad \text{or} \qquad A = A^2BCD = BCD$$

and similarly $B = ACD$, $C = ABD$, and $D = ABC$.

The two-factor interactions are aliased with each other. For example, the alias of AB is CD:

$$AB \cdot I = AB \cdot ABCD \qquad \text{or} \qquad AB = A^2B^2CD = CD$$

The other aliases are $AC = BD$ and $AD = BC$.

The estimates of the main effects and their aliases are found using the four columns of signs in Table For example, from column A we obtain the estimated effect

$$\ell_A = A + BCD = \tfrac{1}{4}(-550 + 749 - 1052 + 650 - 1075 + 642 - 601 + 729)$$
$$=$$

The other columns produce

$$\ell_B = B + ACD = \qquad \ell_C = C + ABD =$$

and

$$\ell_D = D + ABC =$$

Example: Plasma Etch

Clearly, ℓ_A and ℓ_D are large, and if we believe that the three-factor interactions are negligible, the main effects A (gap) and D (power setting) significantly affect etch rate.

The interactions are estimated by forming the AB, AC, and AD columns and adding them to the table. For example, the signs in the AB column are $+, -, -, +, +, -, -, +$, and this column produces the estimate

$$\ell_{AB} = AB + CD = \tfrac{1}{4}(550 - 749 - 1052 + 650 + 1075 - 642 - 601 + 729) =$$

From the AC and AD columns we find

$$\ell_{AC} = AC + BD =$$

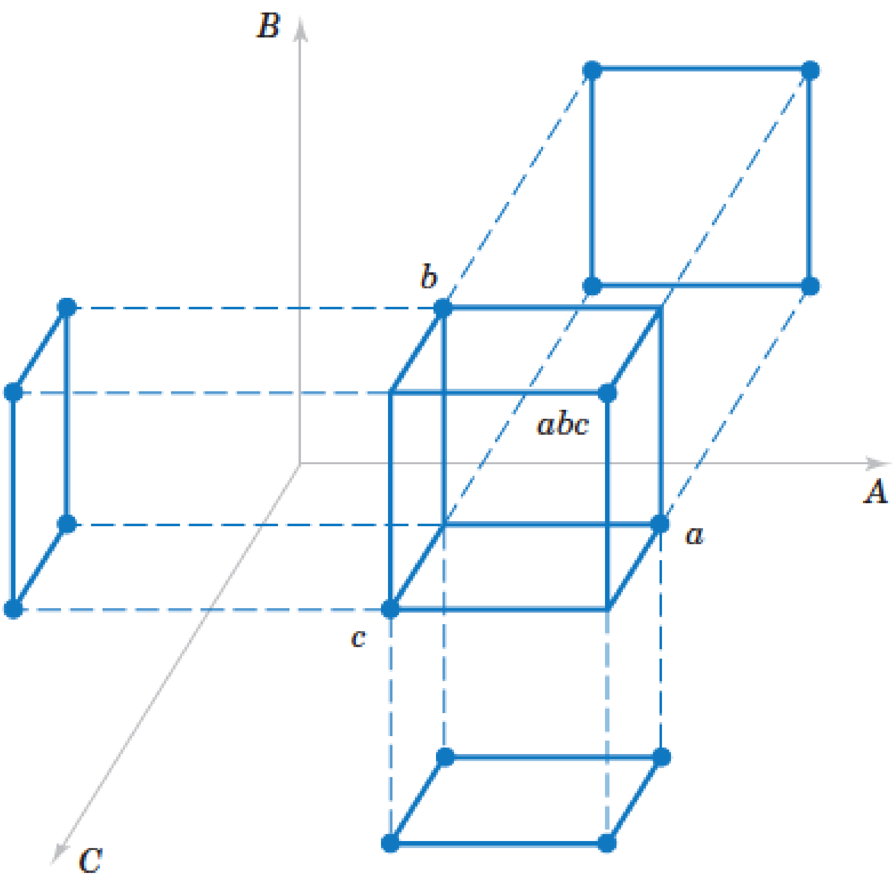
and

$$\ell_{AD} = AD + BC =$$

The ℓ_{AD} estimate is large; the most straightforward interpretation of the results is that since A and D are large, this is the AD interaction. Thus, the results obtained from the 2^{4-1} design agree with the full factorial results in Example 14-5.

Fractional Replication of the 2^k Design

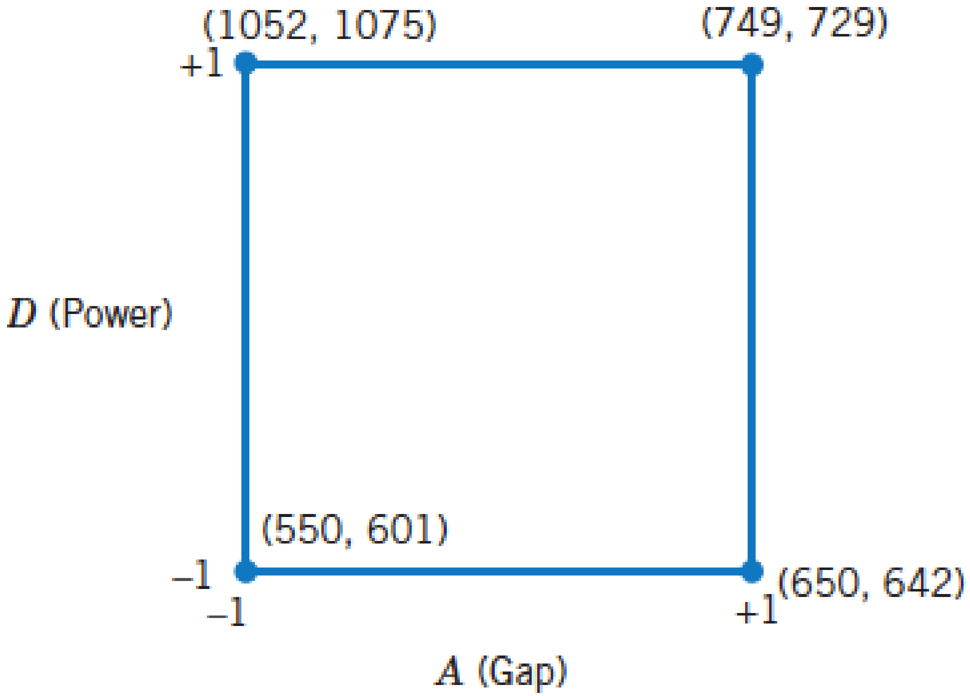
Projection of the 2^{k-1} Design



Projection of a 2^{3-1} design into three 2^2 designs.

Fractional Replication of the 2^k Design

Projection of the 2^{k-1} Design



The 2^2 design obtained by dropping factors *B* and *C* from the plasma etch experiment

Fractional Replication of the 2^k Design

Smaller Fractions: The 2^{k-p} Fractional Factorial

Alias Structure for the 2^{6-2}_{IV} Design with $I = ABCE = BCDF = ADEF$

$A = BCE = DEF = ABCDF$	$AB = CE = ACDF = BDEF$
$B = ACE = CDF = ABDEF$	$AC = BE = ABDF = CDEF$
$C = ABE = BDF = ACDEF$	$AD = EF = BCDE = ABCF$
$D = BCF = AEF = ABCDE$	$AE = BC = DF = ABCDEF$
$E = ABC = ADF = BCDEF$	$AF = DE = BCEF = ABCD$
$F = BCD = ADE = ABCEF$	$BD = CF = ACDE = ABEF$
$ABD = CDE = ACF = BEF$	$BF = CD = ACEF = ABDE$
$ACD = BDE = ABF = CEF$	

Example: Injection Molding

Injection Molding

Parts manufactured in an injection-molding process are showing excessive shrinkage, which is causing problems in assembly operations upstream from the injection-molding area. In an effort to reduce the shrinkage, a quality-improvement team has decided to use a designed experiment to study the injection-molding process. The team investigates six factors—mold temperature (A), screw speed (B), holding time (C), cycle time (D), gate size (E), and holding pressure (F)—each at two levels, with the objective of learning how each factor affects shrinkage and obtaining preliminary information about how the factors interact.

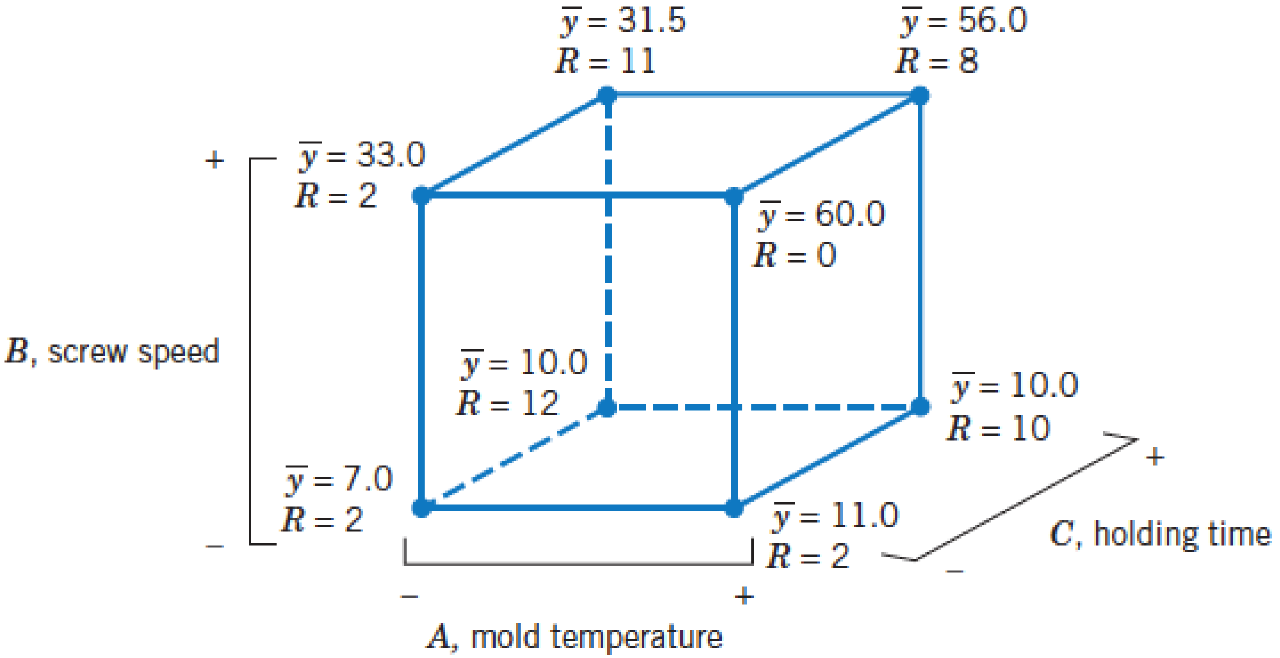
The team decides to use a 16-run two-level fractional factorial design for these six factors. The design is constructed by writing down a 2^4 as the basic design in the factors A, B, C, and D and then setting $E = ABC$ and $F = BCD$ as discussed above. The table shows the design, along with the observed shrinkage ($\times 10$) for the test part produced at each of the 16 runs in the design.

Fractional Replication of the 2^k Design

A 2_{IV}^{6-2} Design for the Injection-Molding Experiment

Run	A	B	C	D	$E = ABC$	$F = BCD$	Observed Shrinkage ($\times 10$)
1	−	−	−	−	−	−	6
2	+	−	−	−	+	−	10
3	−	+	−	−	+	+	32
4	+	+	−	−	−	+	60
5	−	−	+	−	+	+	4
6	+	−	+	−	−	+	15
7	−	+	+	−	−	−	26
8	+	+	+	−	+	−	60
9	−	−	−	+	−	+	8
10	+	−	−	+	+	+	12
11	−	+	−	+	+	−	34
12	+	+	−	+	−	−	60
13	−	−	+	+	+	−	16
14	+	−	+	+	−	−	5
15	−	+	+	+	−	+	37
16	+	+	+	+	+	+	52

Fractional Replication of the 2^k Design



Average shrinkage and range of shrinkage in factors *A*, *B*, and *C*.